

Experimental and Predicted Values of Bubble Point Pressure for Binary and Ternary Systems Consisting of 1-Butanol, 2-Methyl-1-propanol, Glycerol, and Water

Asalil Mustain, Eviana D. Setiawati, Rizky Tetrisyanda, and Gede Wibawa*

Cite This: *J. Chem. Eng. Data* 2022, 67, 941–947

Read Online

ACCESS |



Metrics & More

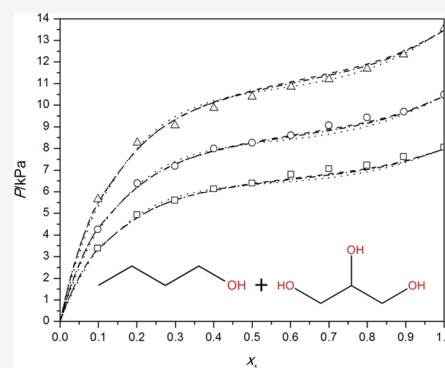


Article Recommendations



Supporting Information

ABSTRACT: In this work, the bubble point pressure data for binary mixtures of (1-butanol + glycerol) and (2-methyl-1-propanol + glycerol) and ternary mixtures of (1-butanol + glycerol + water) and (2-methyl-1-propanol + glycerol + water) were measured using a simple quasi-static ebulliometer. To validate the apparatus, the experimental vapor pressure data of pure 1-butanol and 2-methyl-1-propanol were compared with the literature data and showed average absolute deviations below 0.2%. The measured binary bubble point pressure data in this study show strong positive deviation from ideality based on Raoult's law. For binary systems, the Wilson, nonrandom two-liquid (NRTL), and universal quasi-chemical (UNIQUAC) models were used to correlate the measured experimental data with average absolute deviations in bubble point pressures of 1.6, 1.6, and 2.2% for the 1-butanol + glycerol system and 1.5, 1.6, and 3.0% for the 2-methyl-1-propanol + glycerol system, respectively. Meanwhile, the experimental bubble point pressure data for those ternary systems were compared with the predicted data of Wilson, NRTL, and UNIQUAC models using the binary interaction parameter pair determined from the experimental binary phase equilibrium data of its constituents. The experimental bubble point pressure data for ternary systems were in good agreement with the predicted values using those three activity coefficient models with average absolute deviations in the bubble point pressures within 2.9%.



1. INTRODUCTION

Biodiesel fuel currently receives considerable attention as a sustainable alternative fuel because of biodegradable, nontoxic, renewable, and reducing emissions such as greenhouse gases, volatile organic compounds, particulate matter, and residual hydrocarbons.¹

Biodiesel is a sustainable fuel produced from vegetable oils or animal fats through transesterification reactions, which usually involve alcohols and generate glycerol as a byproduct. One of the factors that obstruct the development of biodiesel is the high cost of production compared to petroleum diesel. While the existing market for glycerol significantly increases, improvement of glycerol quality as a byproduct will make the processing of biodiesel fuel more competitive with petroleum diesel.² Glycerol has advantages in daily necessities such as medical, pharmaceutical, and cosmetics. In food processing industries, glycerol was used as an additive, solvent, and sweetener or a component of packaging materials.³ Glycerol is commonly used with alcohols as mixed solvents in different applications such as a thickener, sweetener, humectant, lubricant, or antifreeze.⁴ In addition, glycerol is also used as a raw material to produce butanol through an acetone–butanol–ethanol (ABE) fermentation process.^{5–8} Due to the wide range of application of glycerol, the phase equilibrium data consisting of glycerol, alcohol, and water mixtures are

extremely important in industry for the development of new technologies and the enhancement of existing processes.

Many binary vapor–liquid equilibrium (VLE) data for systems containing alcohol + glycerol have been published, such as isobaric VLE data of (methanol, ethanol, or 1-propanol) + glycerol systems at various pressures measured by Soujanya et al.,⁹ Coelho et al.,¹⁰ Veneral et al.,¹¹ and Batutah et al.;¹² isobaric VLE data of (methanol, ethanol, 1-propanol, 2-propanol or 1-butanol) + glycerol mixtures at atmospheric pressure reported by Oliveira et al.;³ and isothermal VLE data of ethanol + glycerol systems measured by Zaoui-Djelloul-Daouadji et al.¹³ Moreover, isothermal VLE data of (methanol, ethanol, 1-propanol, or 2-propanol) + glycerol at different temperatures were reported in our previous work.^{14,15} For ternary systems involving alcohol, glycerol, and water, the VLE data for those mixtures were reported in previous studies.^{11,16–22}

Received: December 7, 2021

Accepted: March 21, 2022

Published: April 1, 2022

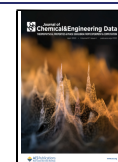


Table 1. Material Specification and Properties of Pure Components

component	CAS reg. no.	source	mass fraction purity	MW (g·mol ⁻¹)	n_D at ($T = 298.15$ K and $P = 98.8$ kPa)	
					exp	lit
1-butanol	71-36-3	Merck Germany	0.995	74.123	1.3974	1.3975 ^b
2-methyl-1-propanol	78-83-1	Merck Germany	0.990	74.123	1.3938	1.3939 ^c
glycerol	56-81-5	Merck Germany	0.995	92.095	1.4723	1.4726 ^d
water ^a	7732-18-5			18.015	1.3325	1.3325 ^d

^aElectrical conductivity $\leq 3.0 \mu\text{S}\cdot\text{cm}^{-1}$ at 298.15 K. ^bRef 36. ^cRef 37. ^dRef 18.

Based on our knowledge, no binary bubble point pressure data of the (1-butanol or 2-methyl-1-propanol) + glycerol systems and ternary bubble point pressure data of the (1-butanol or 2-methyl-1-propanol) + glycerol + water systems have been reported. Therefore, in this work, the bubble point pressure for those systems in the temperature range of 303.15–343.15 K was experimentally measured using a simple quasi-static ebulliometer apparatus. The binary measured data were correlated with Wilson,²³ nonrandom two-liquid (NRTL),²⁴ and universal quasi-chemical (UNIQUAC)²⁵ models. The obtained ternary bubble point pressure data in this work were compared with the predicted values of Wilson, NRTL, and UNIQUAC models using the binary interaction parameter pairs from the binary phase equilibrium data of its constituent substances.

2. EXPERIMENTAL SECTION

2.1. Materials. Organic chemicals such as 1-butanol (purity ≥ 0.995), 2-methyl-1-propanol (purity ≥ 0.990), and glycerol (purity ≥ 0.995) were purchased from Merck (Germany) and used without additional purification. Water was double-distilled for the ternary mixtures. The material specifications are provided in Table 1. The refractive indices were determined with a digital Abbe refractometer (Kruss AR2008, Germany) connected to a liquid circulating thermostatic bath (Ovan, Spain) to control the temperature of the measurement within ± 0.1 K. The estimated refractive index uncertainties were less than 0.0002.

2.2. Apparatus and Procedures. In this study, a simple quasi-static ebulliometer connected with condenser, vacuum, and control temperature units was used to measure the vapor pressures of the pure component and the bubble point pressure of the mixtures. The details and schematic diagram of the ebulliometer were reported in our previous work.²⁶

The procedure of the experiment was conducted by introducing liquid mixtures of known composition, which was gravimetrically determined into an ebulliometer cell. Then, the vacuum pump was switched on to relieve air and impurities in the equilibrium cell and the line package. An electric heater was used to heat the mixtures in the equilibrium cell, and the temperature was adjusted with a controller (Shimaden SR64). The temperature in the equilibrium cell was measured by a four-wire platinum resistance temperature detector and recorded by a YOKOGAWA 7563. During the measurement, the mixture was homogeneously mixed with a magnetic stirrer. The equilibrium condition at the desired temperature was reached, as shown by constant pressure in the system. Thus, the values were recorded as the bubble point pressure. Due to the large volume of the liquid mixtures charged into the ebulliometer cell, the liquid-phase composition under equilibrium condition was assumed to be identical as its initial composition because there is no significant change in the

liquid-phase composition during measurements as proved in our previous work.²⁶ The bubble point pressure measurement was repeated for different temperatures and compositions.

The measurement uncertainties came from the preparation of initial liquid mixtures and pressure and temperature measurements in the equilibrium cell. The initial liquid mixture compositions were prepared gravimetrically by weighing the required amounts of the pure components on an Ohaus balance with an accuracy of ± 0.0002 g. The estimated uncertainty of the liquid-phase mole fraction under equilibrium condition was less than 0.005. The temperature was controlled within ± 0.1 K of the desired value. The bubble point pressures at several temperatures and compositions were measured using a U-shaped mercury manometer with a precision of ± 0.1 kPa.

3. RESULTS AND DISCUSSION

The vapor pressures of pure 1-butanol and 2-methyl-1-propanol measured in this work are presented in Table S1 in the Supporting Information and compared with the calculated values using the Antoine equation. The parameters were collected from the Thermodynamics Research Center (TRC) data bank, College Station, TX, as listed in Table S2 in the Supporting Information.^{27,28} As presented in Figure 1, the comparisons show that the experimental data are consistent with the literature values, and the average absolute deviations in the vapor pressures are less than 0.14 and 0.20%, respectively. The reliability tests of the apparatus in this study were also validated in our previous work.^{29–31}

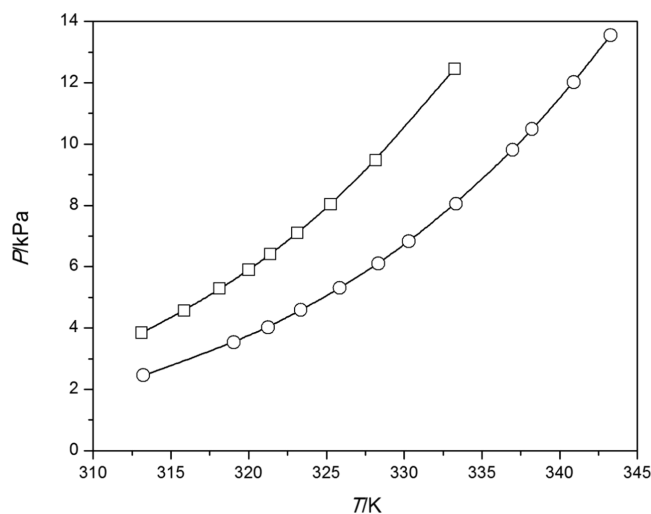


Figure 1. Vapor pressures of the pure components: (○) experimental data of 1-butanol; (□) experimental data of 2-methyl-1-propanol; and (—) calculated values from the Antoine equation.

The apparatus was subsequently used to determine the bubble point pressure for binary systems of 1-butanol (1) + glycerol (2) at temperatures of 333.15–343.15 K and 2-methyl-1-propanol (1) + glycerol (2) at temperatures of 313.15–323.15 K, as presented in Tables 2 and 3, respectively.

Table 2. Bubble Point Pressure Data for the Binary System of 1-Butanol (1) + Glycerol (2) at $T = 333.15$ – 343.15 K^a

x_1	P (kPa)		
	$T = 333.15$ K	$T = 338.15$ K	$T = 343.15$ K
0.0984	3.38	4.25	5.64
0.2011	4.93	6.39	8.26
0.2988	5.59	7.19	9.06
0.4002	6.13	7.99	9.86
0.5000	6.39	8.26	10.39
0.6014	6.79	8.61	10.85
0.7000	7.06	9.06	11.19
0.7996	7.22	9.43	11.67
0.8948	7.62	9.70	12.34
1.0000	8.05	10.48	13.54

^aStandard uncertainties u are $u(T) = 0.1$ K, $u(P) = 0.1$ kPa, and $u(x_1) = 0.005$.

Table 3. Bubble Point Pressure Data for the Binary System of 2-Methyl-1-propanol (1) + Glycerol (2) at $T = 313.15$ – 323.15 K^a

x_1	P (kPa)		
	$T = 313.15$ K	$T = 318.15$ K	$T = 323.15$ K
0.0990	1.99	2.71	3.44
0.2029	2.79	3.59	4.87
0.3020	3.15	4.12	5.80
0.3982	3.33	4.47	6.02
0.4976	3.41	4.55	6.09
0.6022	3.54	4.71	6.21
0.7004	3.61	4.84	6.33
0.7955	3.65	4.98	6.45
0.9005	3.68	5.05	6.57
1.0000	3.83	5.27	7.09

^aStandard uncertainties u are $u(T) = 0.1$ K, $u(P) = 0.1$ kPa, and $u(x_1) = 0.005$.

The Wilson, NRTL, and UNIQUAC models were used to correlate the measured bubble point pressure data for the binary system in this study. The physical properties and parameters of each component used in the binary interaction parameter correlation are given in Table S3 in the Supporting Information.^{27,28,32} Because the equilibrium pressure data were only P – T – x_i data, the optimal values of the binary interaction parameters for the binary mixtures studied in this work were obtained using Barker's method.³³ The following objective function (OF) is selected in the experimental data correlation

$$OF = \sum_{i=1}^n (P_{i,\text{cal}} - P_{i,\text{exp}})^2 \quad (1)$$

where n is the number of data points, and the subscripts cal and exp refer to calculated and experimental values, respectively. Because of the low-pressure system, the vapor phase of the mixture could be assumed to follow the ideal gas behavior. The nonideality of the liquid phase was expressed

with the activity coefficient model. Based on these conditions, the equilibrium pressures are calculated as follows

$$P_{\text{cal}} = \sum_{i=1}^m x_i \gamma_i P_i^s \quad (2)$$

where m is the number of components in the mixtures, x_i is the liquid-phase mole fraction under equilibrium condition, and γ_i is the activity coefficient of the components. P_i^s is the vapor pressure of the pure components calculated using the Antoine equation. The average absolute deviation (AAD) in the bubble point pressure is used to evaluate the quality of the experimental data correlations as follows

$$AAD = \frac{1}{n} \sum_{i=1}^n \left| \frac{P_{i,\text{cal}} - P_{i,\text{exp}}}{P_{i,\text{exp}}} \right| \cdot 100\% \quad (3)$$

From the experimental data correlations, the best fitted binary interaction parameters (A_{12} and A_{21}) for the system in this work were obtained, as shown in Table 4, with the average

Table 4. Optimum Binary Interaction Parameters of the Activity Coefficient Models and Average Absolute Deviations (AADs)

model	parameters			AAD ^a (%)
	A_{12} (J·mol ⁻¹)	A_{21} (J·mol ⁻¹)	α	
1-Butanol (1) + Glycerol (2), $T = 333.15\text{--}343.15$ K				
Wilson ^b	2449.7	4741.5		1.6
NRTL ^c	3811.4	3181.2	0.47	1.6
UNIQUAC ^d	−54.3	1339.4		2.2
2-Methyl-1-propanol (1) + Glycerol (2), $T = 313.15\text{--}323.15$ K				
Wilson	2638.6	6521.5		1.5
NRTL	5320.7	3540.7	0.47	1.6
UNIQUAC	348.1	969.4		3.0
Glycerol (1) + Water (2), $T = 323.15$ K ^e				
Wilson	−643.0	686.0		2.0
NRTL	−2926.7	2458.3	0.47	2.0
UNIQUAC	−989.9	907.8		2.0
1-Butanol (1) + Water (2), $T = 323.22$ K ^f				
Wilson	7476.8	6060.3		2.9
NRTL	1910.6	9945.2	0.4	1.4
UNIQUAC	1643.7	1160.5		2.2
2-Methyl-1-propanol (1) + Water (2), $T = 323.15$ K ^f				
Wilson	6083.8	6274.7		2.2
NRTL	2085.3	9592.9	0.4	1.0
UNIQUAC	2082.7	981.9		1.0

^aAAD = $(1/n) \sum_{i=1}^n |(P_{i,\text{cal}} - P_{i,\text{exp}})/P_{i,\text{exp}}| \cdot 100\%$, where n is the number of data points. ^b $A_{ji} = \lambda_{ji} - \lambda_{ii}$. ^c $A_{ji} = g_{ji} - g_{ii}$. ^d $A_{ji} = u_{ji} - u_{ii}$. ^eRecalculated parameters from the experimental data of ref 35. ^fRecalculated parameters from the experimental data of ref 34.

absolute deviation (AAD) in bubble point pressures. The experimental bubble point pressure data agreed well with the Wilson, NRTL, and UNIQUAC equations with AAD values of 1.6, 1.6, and 2.2% for the 1-butanol + glycerol system and 1.5, 1.6, and 3.0% for the 2-methyl-1-propanol + glycerol system, respectively. To illustrate the correlation results, the experimental data and the corresponding calculated bubble point pressures using the Wilson, NRTL, and UNIQUAC activity coefficient models are plotted in Figures 2 and 3. As an explanation of these figures, the points are the experimental data, and the lines are the calculated values using the Wilson,

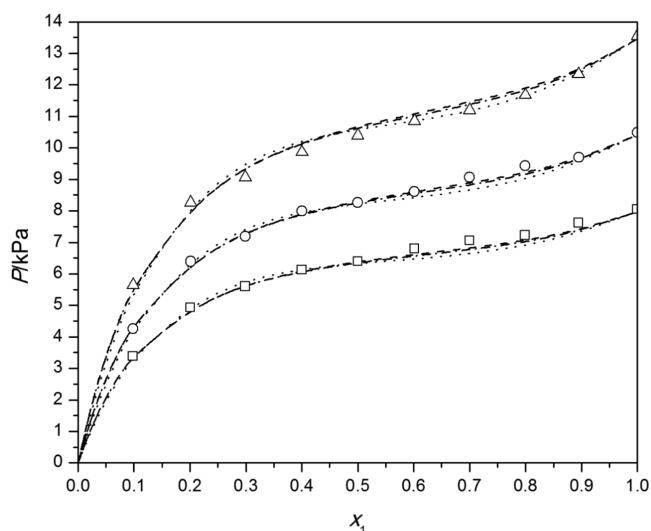


Figure 2. Bubble point pressure (P – x_1) diagram for the binary system of 1-butanol (1) + glycerol (2): (□) experimental data at 333.15 K; (○) experimental data at 338.15 K; (Δ) experimental data at 343.15 K; (---) calculated values from the Wilson model; (–•–) calculated values from the NRTL model; and (•••••) calculated values from the UNIQUAC model.

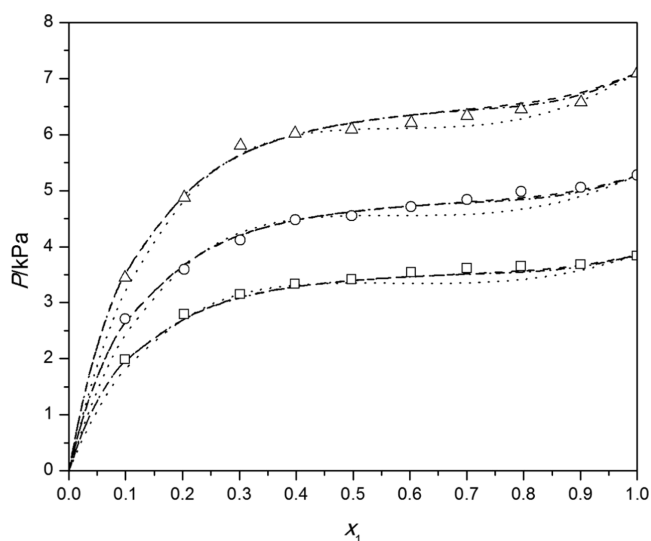


Figure 3. Bubble point pressure (P – x_1) diagram for the binary system of 2-methyl-1-propanol (1) + glycerol (2): (□) experimental data at 313.15 K; (○) experimental data at 318.15 K; (Δ) experimental data at 323.15 K; (---) calculated values from the Wilson model; (–•–) calculated values from the NRTL model; and (•••••) calculated values from the UNIQUAC model.

NRTL, and UNIQUAC models. As illustrated in Figures 2 and 3, the bubble point pressure data for binary mixtures of 1-butanol or 2-methyl-1-propanol with glycerol show a strong positive deviation from Raoult's law. The same trend was shown by the P – x_1 diagram of both systems, the bubble point pressures were increased when the mole fractions of 1-butanol or 2-methyl-1-propanol increase with a significant increase occurred at low 1-butanol or 2-methyl-1-propanol mole fraction.

In this work, a ternary system consisting of 1-butanol, 2-methyl-1-propanol, glycerol, and water was also measured. The experimental bubble point pressure data for the ternary

systems of 1-butanol + glycerol + water at temperatures of 313.15–343.15 K and 2-methyl-1-propanol + glycerol + water at temperatures of 303.15–333.15 K are presented in Tables 5 and 6. These ternary system phase behaviors contain two-phase liquid regions at certain compositions because 1-butanol and 2-methyl-1-propanol are partially miscible with water. Therefore, before the bubble point pressure measurements were performed, it was important to comprehend the total miscibility region of those mixtures to evade the two-layer liquid-phase formation. For this purpose, the binary phase equilibrium data of C_4 -alcohols + water mixtures published by Fischer and Gmehling³⁴ were used as the basis to determine liquid mixtures of the ternary systems. The two-layer liquid-phase region for the binary system of 1-butanol (1) + water (2) at $T = 232.23$ K and 2-methyl-1-propanol (1) + water (2) at $T = 323.15$ K were in the range of $x_1 = 0.01495$ – 0.48563 and 0.01789 – 0.54237 , respectively.

For the ternary systems, the experimental bubble point pressure data obtained in this work were compared with the predictions made using the Wilson, NRTL, and UNIQUAC models. The binary interaction parameters of those equations were determined from the experimental phase equilibrium data of its constituent binary systems. The binary interaction parameters of glycerol + water were correlated from Dulitskaya.³⁵ The experimental data from Fischer and Gmehling³⁴ were also used in the binary interaction parameter determination for (1-butanol or 2-methyl-1-propanol) + water systems. The binary interaction parameters (A_{12} and A_{21}) in the prediction for those ternary systems are listed in Table 4.

In addition, the ternary bubble point pressure data were evaluated using the average absolute deviation (AAD), as described in eq 3. Table 7 summarizes the comparison result of bubble point pressures between the experimental data and the prediction values of Wilson, NRTL, and UNIQUAC models. The experimental data were found to be compatible with the predicted values. The average absolute deviations (AADs) between the experimental data and predicted values using Wilson, NRTL, and UNIQUAC equations in the bubble point pressures were less than 2.9%.

As an illustration, Figure 4 compares the experimental bubble point pressure data and predicted values using Wilson, NRTL, and UNIQUAC models for the ternary systems of 1-butanol (1) + glycerol (2) + water (3) and 2-methyl-1-propanol (1) + glycerol (2) + water (3), respectively. These predictions are calculated using only binary interaction parameters of the related binary systems. The activity coefficient models can satisfactorily describe the ternary systems, although the application of the binary interaction parameters in this study is extrapolated from their temperature measurement conditions. At the similar composition as illustrated in Figure 4, the bubble point pressures of 1-butanol (1) + glycerol (2) + water (3) are lower than those of 2-methyl-1-propanol (1) + glycerol (2) + water (3) caused by the lower vapor pressures of 1-butanol compared to 2-methyl-1-propanol. Both compounds have the same molecular weight with different molecular structures. This condition occurs because all four carbon atoms of 1-butanol are lined up in a straight line, while 2-methyl-1-propanol has one carbon atom branching off from the main chain. The reliable model development will be useful for the simulation, design, and optimization of various separation processes in industries.

Table 5. Bubble Point Pressure Data for the Ternary System of 1-Butanol (1) + Glycerol (2) + Water (3) at $T = 313.15$ – 343.15 K^a

x_1	x_2	P (kPa)						
		$T = 313.15$ K	$T = 318.15$ K	$T = 323.15$ K	$T = 328.15$ K	$T = 333.15$ K	$T = 338.15$ K	$T = 343.15$ K
0.0036	0.3000	4.85	6.35	8.13	10.19	12.93	16.22	20.64
0.0100	0.3000	5.13	6.61	8.57	11.07	13.93	17.58	22.14
0.0700	0.3000	6.01	7.85	10.09	13.03	16.81	20.97	26.97
0.3500	0.3000	4.98	6.69	8.87	11.19	14.49	17.94	23.37
0.3800	0.3000	4.75	6.41	8.57	10.81	14.05	17.68	22.35
0.4498	0.3003	4.32	5.68	7.71	9.97	12.43	16.27	20.43
0.5000	0.3000	3.91	5.24	7.16	9.09	11.96	14.83	19.11
0.5704	0.2997	3.39	4.60	5.94	7.81	10.06	13.06	16.74
0.6205	0.2996	2.91	4.00	5.10	7.07	9.07	11.90	14.46
0.6800	0.3000	2.43	3.22	4.47	5.63	7.32	9.54	12.26

^aStandard uncertainties u are $u(T) = 0.1$ K, $u(P) = 0.1$ kPa, and $u(x_i) = 0.005$.**Table 6. Bubble Point Pressure Data for the Ternary System of 2-Methyl-1-propanol (1) + Glycerol (2) + Water (3) at $T = 303.15$ – 333.15 K^a**

x_1	x_2	P (kPa)						
		$T = 303.15$ K	$T = 308.15$ K	$T = 313.15$ K	$T = 318.15$ K	$T = 323.15$ K	$T = 328.15$ K	$T = 333.15$ K
0.0016	0.3000	2.79	3.77	4.68	6.41	8.27	10.67	13.19
0.0090	0.3000	2.92	3.93	5.14	6.74	8.70	11.15	14.15
0.5400	0.3000	2.61	3.54	5.01	6.47	8.87	11.54	14.74
0.6600	0.3000	1.95	2.74	3.77	5.11	6.83	9.02	11.78
0.6825	0.2977	1.85	2.62	3.59	4.87	6.59	8.64	11.33
0.0014	0.4000	2.21	2.93	3.87	5.04	6.50	8.35	10.51
0.0026	0.4000	2.23	2.97	3.93	5.12	6.64	8.40	10.67
0.0077	0.4000	2.32	3.09	4.08	5.33	6.96	9.39	11.52
0.4886	0.4000	2.27	3.17	4.37	5.84	7.71	9.97	13.17
0.5314	0.4000	2.00	2.91	4.11	5.57	7.44	9.71	12.59
0.5657	0.4000	1.92	2.77	3.84	5.31	7.17	9.44	12.32
0.0012	0.5000	1.76	2.35	3.08	4.01	5.17	6.51	8.37
0.0021	0.5000	1.76	2.35	3.08	4.01	5.17	6.77	8.91
0.0064	0.5000	1.87	2.51	3.28	4.28	5.71	7.04	9.17
0.4072	0.5000	2.06	2.86	3.92	5.29	7.04	9.25	12.02
0.4428	0.5000	1.95	2.72	3.74	5.06	6.75	8.90	11.58
0.4714	0.5000	1.86	2.61	3.59	4.87	6.51	8.59	11.21
0.4859	0.5000	1.82	2.55	3.51	4.77	6.39	8.44	11.02

^aStandard uncertainties u are $u(T) = 0.1$ K, $u(P) = 0.1$ kPa, and $u(x_i) = 0.005$.**Table 7. Absolute Average Deviation (AAD) in Bubble Point Pressures from the Prediction Values Using Wilson, NRTL, and UNIQUAC Models**

system	T (K)	n^a	model	AAD ^a (%)
1-butanol (1) + glycerol (2) + water (3)	313.15–343.15	70	Wilson	2.0
			NRTL	1.8
			UNIQUAC	2.9
2-methyl-1-propanol (1) + glycerol (2) + water (3)	303.15–333.15	126	Wilson	1.9
			NRTL	2.2
			UNIQUAC	1.7

^aAAD = $(1/n) \sum_{i=1}^n |(P_{\text{cal}} - P_{\text{exp}})/P_{\text{exp}}| \cdot 100\%$, where n is the number of data points.

4. CONCLUSIONS

In this study, the bubble point pressure data for binary 1-butanol or 2-methyl-1-propanol (1) + glycerol (2) and ternary 1-butanol or 2-methyl-1-propanol (1) + glycerol (2) + water (3) systems were conducted using a simple quasi-static ebulliometer at various temperatures. For the binary experimental data correlations, the Wilson, NRTL, and UNIQUAC models gave good correlation results with average

absolute deviations of 1.6, 1.6, and 2.2% for the 1-butanol (1) + glycerol (2) system and 1.5, 1.6, and 3.0% for the 2-methyl-1-propanol (1) + glycerol (2) system, respectively. For the ternary systems, the experimental bubble point pressures were compared with the prediction values made using the Wilson, NRTL, and UNIQUAC models with the binary parameters of its constituent binary systems. The measured ternary bubble point pressure data were compatible with the predicted values with average absolute deviations below 2.9%. The experimental

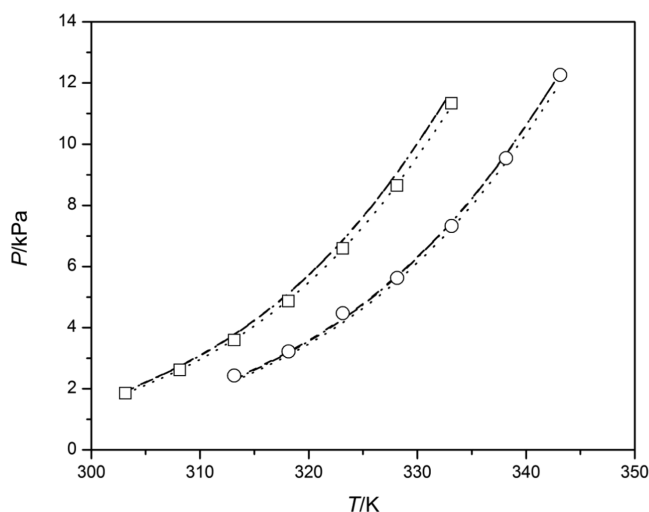


Figure 4. Bubble point pressures for the ternary system: (○) experimental data of 1-butanol (1) + glycerol (2) + water (3) at $x_1 = 0.6800$ and $x_2 = 0.3000$; (□) experimental data of 2-methyl-1-propanol (1) + glycerol (2) + water (3) at $x_1 = 0.6825$ and $x_2 = 0.2997$; (---) calculated values from the Wilson model; (—●—) calculated values from the NRTL model; and (●●●●) calculated values from the UNIQUAC model.

bubble point pressure data and correlation obtained in this work will be beneficial for the simulation, design, and optimization of various processes in industries consisting of glycerol, butanol, and water systems.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jced.1c00937>.

Vapor pressures of 1-butanol and 2-methyl-1-propanol; parameters of the Antoine equation for pure compounds; and physical properties and parameters of pure components in the activity coefficient correlation (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Gede Wibawa – Department of Chemical Engineering, Faculty of Industrial Technology and Systems Engineering, Sepuluh Nopember Institute of Technology (ITS), Kampus ITS Sukolilo, Surabaya 60111, Indonesia; orcid.org/0000-0002-1255-9210; Phone: +62-31-5946240; Email: gwibawa@chem-eng.its.ac.id; Fax: +62-31-5999282

Authors

Asail Mustain – Department of Chemical Engineering, Faculty of Industrial Technology and Systems Engineering, Sepuluh Nopember Institute of Technology (ITS), Kampus ITS Sukolilo, Surabaya 60111, Indonesia; Department of Chemical Engineering, State Polytechnic of Malang, Malang 65141, Indonesia; orcid.org/0000-0002-3527-3928

Eviana D. Setiawati – Department of Chemical Engineering, Faculty of Industrial Technology and Systems Engineering, Sepuluh Nopember Institute of Technology (ITS), Kampus ITS Sukolilo, Surabaya 60111, Indonesia

Rizky Tetrisyanda – Department of Chemical Engineering, Faculty of Industrial Technology and Systems Engineering,

Sepuluh Nopember Institute of Technology (ITS), Kampus ITS Sukolilo, Surabaya 60111, Indonesia

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.jced.1c00937>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was supported by World Class Research (WCR) research grants from Direktorat Riset dan Pengabdian Masyarakat, Institut Teknologi Sepuluh Nopember, Contract No: 1043/PKS/ITS/2021. The authors would like to express their appreciation to Ajeng P. Yudiputri, Randhi R. Agustin, and Risha R. Suminar for their help in the experimental study.

■ REFERENCES

- (1) Mesquita, F. M. R.; Evangelista, N. S.; de Sant'Ana, H. B.; de Santiago-Aguiar, R. S. Liquid–Liquid Equilibrium for the Glycerol + Alcohol + Coconut Biodiesel System at Different Temperatures and Atmospheric Pressure. *J. Chem. Eng. Data* **2012**, *57*, 3557–3562.
- (2) Keera, S. T.; El Sabagh, S. M.; Taman, A. R. Transesterification of Vegetable Oil to Biodiesel Fuel Using Alkaline Catalyst. *Fuel* **2011**, *90*, 42–47.
- (3) Oliveira, M. B.; Teles, A. R. R.; Queimada, A. J.; Coutinho, J. A. P. Phase Equilibria of Glycerol Containing Systems and Their Description with the Cubic-Plus-Association (CPA) Equation of State. *Fluid Phase Equilib.* **2009**, *280*, 22–29.
- (4) Chabouni, Y.; Amireche, F. Some Physicochemical Properties of Binary Mixtures of Glycerol with Butanol Isomers in the Temperature Range 293.15–318.15 K and Ambient Pressure. *J. Chem. Eng. Data* **2020**, *65*, 1679–1694.
- (5) Lin, D.-S.; Yen, H.-W.; Kao, W.-C.; Cheng, C.-L.; Chen, W.-M.; Huang, C.-C.; Chang, J.-S. Bio-butanol production from glycerol with *Clostridium pasteurianum* CH4: the effects of butyrate addition and in situ butanol removal via membrane distillation. *Biotechnol. Biofuels* **2015**, *8*, No. 168.
- (6) Saini, M.; Lin, L.-J.; Chiang, C.-J.; Chao, Y.-P. Effective production of n-butanol in *Escherichia coli* utilizing the glucose–glycerol mixture. *J. Taiwan Inst. Chem. Eng.* **2017**, *81*, 134–139.
- (7) Chen, T.; Xu, F.; Zhang, W.; Zhou, J.; Dong, W.; Jiang, Y.; Lu, J.; Fang, Y.; Jiang, M.; Xin, F. High butanol production from glycerol by using *Clostridium* sp. strain CT7 integrated with membrane assisted pervaporation. *Bioresour. Technol.* **2019**, *288*, No. 121530.
- (8) Sarchami, T.; Rehmann, L. Increased Butanol Yields through Cosubstrate Fermentation of Jerusalem Artichoke Tubers and Crude Glycerol by *Clostridium pasteurianum* DSM 525. *ACS Omega* **2019**, *4*, 15521–15529.
- (9) Soujanya, J.; Satyavathi, B.; Vittal Prasad, T. E. Experimental (Vapour + Liquid) Equilibrium Data of (Methanol + Water), (Water + Glycerol) and (Methanol + Glycerol) Systems at Atmospheric and Sub-Atmospheric Pressures. *J. Chem. Thermodyn.* **2010**, *42*, 621–624.
- (10) Coelho, R.; dos Santos, P. G.; Mafra, M. R.; Cardozo-Filho, L.; Corazza, M. L. (Vapor + Liquid) Equilibrium for the Binary Systems {Water + Glycerol} and {Ethanol + Glycerol, Ethyl Stearate, and Ethyl Palmitate} at Low Pressures. *J. Chem. Thermodyn.* **2011**, *43*, 1870–1876.
- (11) Venerál, J. G.; Benazzi, T.; Mazutti, M. A.; Voll, F. A. P.; Cardozo-Filho, L.; Corazza, M. L.; Guirardello, R.; Oliveira, J. V. Thermophysical Properties of Biodiesel and Related Systems. Part I. Vapour–Liquid Equilibrium at Low Pressures of Binary and Ternary Systems Involving Methanol, Ethanol, Glycerol, Water and NaCl. *J. Chem. Thermodyn.* **2013**, *58*, 398–404.
- (12) Batutah, M. A.; Kuswandi, K.; Wibawa, G. Isobaric Binary Vapor–Liquid Equilibrium of Ethanol + Glycerol and 1-Propanol + Glycerol Systems at 16.0 and 101.3 kPa. *J. Chem. Eng. Data* **2020**, *65*, 3802–3807.

- (13) Zaoui-Djelloul-Daouadi, M.; Negadi, A.; Mokbel, I.; Negadi, L. (Vapor-Liquid) Equilibria and Excess Gibbs Free Energy Functions of (Ethanol + Glycerol), or (Water + Glycerol) Binary Mixtures at Several Temperatures. *J. Chem. Thermodyn.* **2014**, *69*, 165–171.
- (14) Wibawa, G.; Mustain, A.; Akbarina, M. F.; Ruslim, R. M. Isothermal Vapor–Liquid Equilibrium of Ethanol + Glycerol and 2-Propanol + Glycerol at Different Temperatures. *J. Chem. Eng. Data* **2015**, *60*, 955–959.
- (15) Wiguno, A.; Mustain, A.; Irwansyah, W. F. E.; Wibawa, G. Isothermal Vapor–Liquid Equilibrium of Methanol + Glycerol and 1-Propanol + Glycerol. *Indones. J. Chem.* **2016**, *16*, 111–116.
- (16) Soujanya, J.; Satyavathi, B.; Sankarshana, T. Experimental and Predicted (Vapor + Liquid) Equilibrium Data of Ternary Mixtures of Methanol, Water, and Propan-1,2,3-triol at 96 kPa. *J. Chem. Eng. Data* **2011**, *56*, 3309–3311.
- (17) Verhoeve, L. A. J.; Lauwers, E. Vapor-Liquid Equilibrium of the System 2-Propanol-Water-1,2,3-Propanetriol at 760 Mm. of Hg. *J. Chem. Eng. Data* **1969**, *14*, 306–309.
- (18) Pla-Franco, J.; Lladosa, E.; Loras, S.; Montón, J. B. Phase Equilibria for the Ternary Systems Ethanol, Water + Ethylene Glycol or + Glycerol at 101.3 kPa. *Fluid Phase Equilib.* **2013**, *341*, 54–60.
- (19) Souza, W. L. R.; Silva, C. S.; Meleiro, L. A. C.; Mendes, M. F. Vapor–liquid equilibrium of the (water + ethanol + glycerol) system: Experimental and modelling data at normal pressure. *J. Chem. Thermodyn.* **2013**, *67*, 106–111.
- (20) Zhang, L.; Zhang, W.; Yang, B. Experimental Measurement and Modeling of Ternary Vapor–Liquid Equilibrium for Water + 2-Propanol + Glycerol. *J. Chem. Eng. Data* **2014**, *59*, 3825–3830.
- (21) Zhang, L.; Yang, B.; Zhang, W. Vapor–Liquid Equilibrium of Water + Ethanol + Glycerol: Experimental Measurement and Modeling for Ethanol Dehydration by Extractive Distillation. *J. Chem. Eng. Data* **2015**, *60*, 1892–1899.
- (22) Zhang, L.; Wang, W.; Zhu, X.; Zhang, Z. Experimental Measurement and Modeling of Vapor–Liquid Equilibrium for the Ternary System Water + 1-Propanol + Glycerol. *J. Chem. Eng. Data* **2016**, *61*, 1637.
- (23) Wilson, G. M. Vapor-Liquid Equilibrium. XI. A New Expression for the Excess Free Energy of Mixing. *J. Am. Chem. Soc.* **1964**, *86*, 127–130.
- (24) Renon, H.; Prausnitz, J. M. Local Compositions in Thermodynamic Excess Functions for Liquid Mixtures. *AIChE J.* **1968**, *14*, 135–144.
- (25) Abrams, D. S.; Prausnitz, J. M. Statistical Thermodynamics of Liquid Mixtures: A New Expression for the Excess Gibbs Energy of Partly or Completely Miscible Systems. *AIChE J.* **1975**, *21*, 116–128.
- (26) Oktavian, R.; Amidelsi, V.; Rasmito, A.; Wibawa, G. Vapor Pressure Measurements of Ethanol–Isooctane and 1-Butanol–Isooctane Systems Using a New Ebulliometer. *Fuel* **2013**, *107*, 47–51.
- (27) Poling, B. E.; Prausnitz, J. M.; O'Connell, J. P. *The Properties of Gases and Liquids*, 5th ed.; McGraw-Hill: New York, 2001.
- (28) Reid, R. C.; Prausnitz, J. M.; Poling, B. E. *The Properties of Gases and Liquids*, 4th ed.; McGraw-Hill: New York, 1987.
- (29) Anugraha, R. P.; Altway, A.; Wibawa, G. Measurement and Correlation of Isothermal Binary Vapor–Liquid Equilibrium for Diethyl Carbonate + Isooctane/n-Heptane/Toluene Systems. *J. Chem. Eng. Data* **2017**, *62*, 2362–2366.
- (30) Anugraha, R. P.; Wiguno, A.; Altway, A.; Wibawa, G. Vapor pressures of diethyl carbonate + ethanol binary mixture and diethyl carbonate + ethanol + isooctane/toluene ternary mixtures at temperatures range of 303.15–323.15 K. *J. Mol. Liq.* **2018**, *264*, 32–37.
- (31) Wiguno, A.; Tetrisyanda, R.; Wibawa, G. Vapor Pressure of 2-Butanol + Diethyl Carbonate and tert-Butanol + Diethyl Carbonate at the Temperature of 303.15–323.15 K. *J. Chem. Eng. Data* **2020**, *65*, 2441–2445.
- (32) Hansen, H. K.; Rasmussen, P.; Fredenslund, A.; Schiller, M.; Gmehling, J. Vapor-liquid equilibria by UNIFAC group contribution. 5. Revision and extension. *Ind. Eng. Chem. Res.* **1991**, *30*, 2352–2355.
- (33) Barker, J. A. Determination of Activity Coefficients from Total Pressure Measurements. *Aust. J. Chem.* **1953**, *6*, 207–210.
- (34) Fischer, K.; Gmehling, J. P - x and γ^∞ Data for the Different Binary Butanol-Water Systems at 50°C. *J. Chem. Eng. Data* **1994**, *39*, 309–315.
- (35) Dulitskaya, K. A. Vapor Pressure of Binary System: I. *Zh. Obshch. Khim.* **1945**, *15*, 9–21.
- (36) Zhang, Y.; Guo, X.; Xu, J.; Wu, Y.; Lu, M. Liquid–Liquid Equilibrium for Ternary Systems, Water + 5-Hydroxymethylfurfural + (1-Butanol, Isobutanol, Methyl Isobutyl Ketone), at 313.15, 323.15, and 333.15 K. *J. Chem. Eng. Data* **2018**, *63*, 2775–2782.
- (37) Chen, S.; Lei, Q.; Fang, W. Density and Refractive Index at 298.15 K and Vapor–Liquid Equilibria at 101.3 kPa for Four Binary Systems of Methanol, n-Propanol, n-Butanol, or Isobutanol with N-Methylpiperazine. *J. Chem. Eng. Data* **2002**, *47*, 811–815.